Siyan Wang

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EDUCATION

University of Connecticut (UConn) Storr, CT
Ph.D Candidate of Statistics *09/2024–current*

- Interest: Extreme value, Risk control, Logistic Regression
- Skills: Statistical computing, Applied Statistics, Optimization

University of Wisconsin–Madison (UW–Madison) Madison, WI
Master of Statistics and Data Science (MSDS) *09/2022–05/2024*

- Skills: Statistical Learning, Deep Learning, Statistical Visualization, Survival Analysis

Huazhong Agricultural University (HZAU) Wuhan, China
Bachelor of Science in Information and Computation Science *09/2018–06/2022*

- Skills: Mathematical Analysis, Machine Learning, Data Mining

RESEARCH EXPERIENCE

Research Assistant, University of Connecticut 09/2024–current
Student Success Analysis
Office of Vice president for Student Life and Enrollment

- Developed predictive modeling pipelines to identify students at elevated risk of DFW outcomes by using selected and created variables.
- Constructed multi-level academic indicators and scenario-based evaluation workflows to examine risk heterogeneity across students, programs, and course contexts.
- Designed visualization and reporting workflows to support institutional decision-making on student success and course-level intervention strategies.

Research Assistant, HZAU 10/2019–05/2022
Study on the Characteristics of Medical Soft Tissue Based on Fractal Dimension
College of Science

- Led a five-member research team on quantitative CT image analysis for lung disease differentiation, including COVID-19 and lung cancer.
- Designed a fractal-dimension-based feature extraction pipeline for 10,000+ CT images across whole-lung and lung-parenchyma regions.
- Benchmarked supervised and unsupervised learning models, with SVM achieving the strongest diagnostic performance at high accuracy.

Research Assistant, HZAU 12/2020–09/2021
Study on Drought Stress of Fruit and Vegetable Seedlings
College of Engineering

- Investigated whether plants lacked water based on outcomes including leaf color, shape, and height.
- Collected depth images and RGB images from different angles using a Kinect tool and infrared camera.
- Calculated the FD of each image and studied the relationship between water content and FD.

PUBLICATION

- Jingfang Shen, Sijie Cheng, **Siyan Wang**, and Wenwei Liu. "A Multi-Scale Stiffness Fractal Model of Joint Interfaces." *Russian Physical Journal*, 2019.

RESEARCH PROJECTS

Body Fat Prediction Analysis in Different Age Groups 10/2023
Course Project, UW–Madison

- Built predictive models for estimating body fat using diverse physiological features and analyzed the distribution of body fat across different age groups.
- Revamped a Shiny app to visually showcase prediction results and create an engaging interactive website for prospective users.
- Contributed to the concept and execution of the age-group analysis and Shiny app construction.

Fruit Classification Based on Various Models and Activation Functions 05/2023
Course Project, UW–Madison

- Implemented multilayer perceptron (MLP) and convolutional neural network (CNN) models for classification on a dataset of 10 fruit classes.
- Built a CNN model by adapting AlexNet to fit fruit image classification.
- Fine-tuned CNN parameters and optimized hyperparameters, including kernel sizes, pooling layers, and activation functions.
- Built and modified models and compared their classification efficiency.

Application of Multidimensional Scaling Algorithm (MDS) 09/2020–01/2021
Course Paper, HZAU

- Verified the feasibility of MDS in data visualization by displaying the relative positions of points on a two-dimensional coordinate plane based on the distance matrix between U.S. cities.
- Compared the MDS visualization results with the cities' actual locations on a map.
- Explored two-dimensional images of outlier data after MDS dimensionality reduction and concluded that MDS visualization becomes less reliable under partial outliers.

AWARDS & PRIZES

- National Scholarship, HZAU (Top 1%), 2019.
- Excellent Student, HZAU (Top 10%), 2019, 2020, 2021, and 2022.

TRAINING

University of Oxford, Exeter College 08/2019
Global Leadership Programme

- Proposed a company acquisition programme and won first place in the final project presentation.
- Studied International Economics, Game Theory, and Globalization.